MIL-A-9482 (USAF)
2 September 1954
Superseding
No. 40395-C
15 December 1947

MILITARY SPECIFICATION

ANTI-ICING EQUIPMENT FOR AIRCRAFT, HEATED SURFACE TYPE,
GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the requirements for airframe thermal anti-icing systems, excluding engine-air-induction systems and radomes.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and publication, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-H-5484	Heaters; Aircraft, Combustion-Type
MIL-T-5842	Transparent Areas, Anti-Icing, Defrosting and Defogging Systems, General Specification For
MIL-E-7563	Electrical Equipment, Aircraft, Installation of, General Specification For
MIL-D-9402	Design of Electrical Equipment in Aircraft, General Specification For

PUBLICATION

AIR FORCE-NAVY AERONAUTICAL BULLETIN

143	Specifications and Standards; Use of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material:

3.1.1 Specifications and standards.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin No. 143, except as provided in the following paragraph:

3.1.1.1 Standard parts.- AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers.

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Commercial utility parts, such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2 Areas to be protected:

3.2.1 All portions of the wing(s) and stabilizing surfaces which are exposed to ice accumulations during flight under the conditions specified in 3.3.1 and 3.3.2 shall be protected by the anti-icing system.

3.2.2 Protection of antenna masts, hinge fairings, spoilers, dive brakes, struts, and other miscellaneous ice accreting parts shall depend upon the effects of: (1) ice accumulation on the normal functioning of the individual part, and (2) increased drag on aircraft performance. These data shall be obtained, evaluated, and submitted to the procuring activity with the contractor's recommendations. The decision on ice protection of these components shall be made by only the procuring activity.

3.2.3 Transparent-area anti-icing shall be in accordance with Specification MIL-T-5842.

3.2.4 The entrance to each air scoop which must function during icing conditions shall be protected against ice accumulations. Protection shall also be provided for the guide vanes or at abrupt changes in direction of ducts leading from these scoops to permit unrestricted airflow in these ducts.

3.3 Design conditions:

3.3.1 The design meteorological icing condition shall be as follows:

(1) Liquid water content 0.5 grams per cubic meter (3.12 by 10^{-5} pounds per cubic foot)
(2) Mean effective droplet diameter - 20 microns (6.56 by 10^{-5} feet)
(3) Altitude - 0 to 20,000 feet

3.3.2 The systems shall be capable of safe operation up to the service ceiling of the aircraft. System capacities shall be based on the following design flight conditions:

(1) Climb at speed for maximum rate of climb
(2) Cruise at speed for maximum range at normal operating altitude, if the normal operating altitude of the airplane is less than 20,000 feet
(3) Descent at speeds, rate of descent, and engine power as recommended by the manufacturer, subject to approval of the procuring activity

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3.3.2.1 If the design mission of the airplane requires intermittent operation below 20,000 feet (e.g., low level reconnaissance, attack, or refueling) typical conditions including the high and low extremes of speed shall be included as design points.

3.3.3 For speeds less than curve A of figure 1, the anti-icing system shall provide complete evaporation of all impinging water droplets at 15°F ambient temperature unless the entire object is heated. In this case a running-wet surface is allowable with a minimum surface temperature of 35°F at 0°F ambient temperature.

3.3.4 For speeds greater than curve A but less than curve B of figure 1, the anti-icing system need only provide a running-wet surface at 0°F ambient temperature since runback will not freeze.

3.3.5 If the design speeds of 3.3.2 are greater than curve B of figure 1, no anti-icing system is required except for landing and takeoff. For this special case the contractor shall submit his recommendations regarding the necessity of an anti-icing system to the procuring activity for a decision.

3.4 Methods of computing heat required and available:

3.4.1 Amount and chordwise extent of water droplet impingement shall be determined by droplet trajectory studies for the airfoil sections used and corrected for the sweepback angle.

3.4.2 External heat transfer coefficients shall be based on the transition from laminar to turbulent flow at the aft edge of the impingement area, or, at a more forward location if so indicated by the airfoil data.

3.4.3 The impingement-area surface shall be considered to be completely wet while areas aft of the impingement area shall be considered to be 20 percent wet.

3.4.4 Water evaporation shall be determined from integration of point evaporation rates and shall not be based on average skin temperatures.

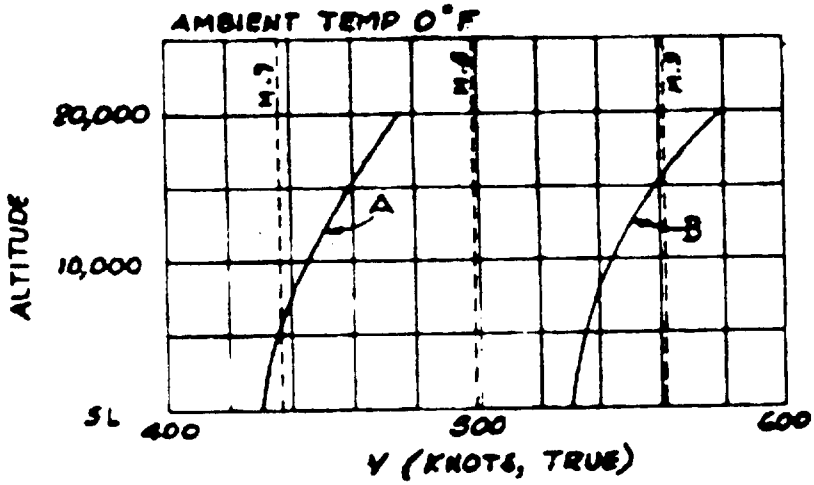
3.4.5 Heat requirements shall be determined at the root, tip, and at least one intermediate section of the wing and at least the root and tip section of each stabilizing surface.

3.4.6 In determining the predicted air temperature available for anti-icing at any wing station, due allowance shall be made for temperature losses involved in airflow from the heat source to that station. Reduction of insulation effectiveness due to crushing, service abuse, and air conduction through duct supports and other heat-loss paths shall be accounted for.

3.4.7 In determining the heat transfer to the outer skin, due allowance shall be made for the effectiveness of the inner skin and the resistance to heat flow from the inner skin to the outer surface.

3.4.8 In determining the heat available, due allowance shall be made for system leakage.

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CURVE A - SPEED ABOVE WHICH RUNBACK DOES NOT FREEZE

CURVE B - SPEED ABOVE WHICH ANTI-ICING SYSTEMS ARE NOT REQUIRED BY THIS SPECIFICATION

NOTE - CURVE "A" IS FOR AN ALUMINUM ALLOY SKIN AFT OF THE HEATED AREA. IF THE CONDUCTIVITY OF THE SKIN IS APPRECIABLY DIFFERENT FROM ALUMINUM ALLOY CURVE "A" CAN NOT BE USED AND THE SPEED AT WHICH RUNBACK DOES NOT FREEZE MUST BE CALCULATED FOR THE PARTICULAR CONSTRUCTION USED.

EFFECT OF SPEED ON ICING REQUIREMENTS

FIGURE 1

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3.5 Heat sources:

3.5.1 The primary heat source shall be hot air obtained from compressor bleed, exhaust-gas heat exchangers, mixtures of compressor bleed air and exhaust gases, or combustion heaters. Electrical heating may be used for small areas where it is more convenient or where it induces less penalty to the airplane performance.

3.5.2 Scoops and ducts leading to exhaust-gas heat exchangers or combustion heaters which provide heat for anti-icing shall be electrically heated.

3.5.3 When more than a single source of heated air is used to furnish heat for the anti-icing system, the several sources shall be manifolded together so that in event of failure of one heat source the remaining heat supply can be distributed to all the surfaces to be anti-iced.

3.6 Detail design:

3.6.1 When any aluminum alloy structural members are heated above 200°F by the anti-icing system, due allowance shall be made for the reduction in allowable stress. In the event that the primary structural members of aluminum alloy are to be heated above 250°F, specific approval of the procuring activity shall be required.

3.6.2 In the event that the heating air contains exhaust products, all ducting, leading edge skins, double skins, et cetera, which the heating air may contact shall be stainless steel or equally corrosion-resistant material.

3.6.3 Distribution ducts shall be insulated to reduce heat losses. Specific requirements for spanwise ducts shall be as shown on figure 2.

3.6.4 The system shall be capable of operation on the ground to assist in ice, frost, and snow removal from the airframe surfaces.

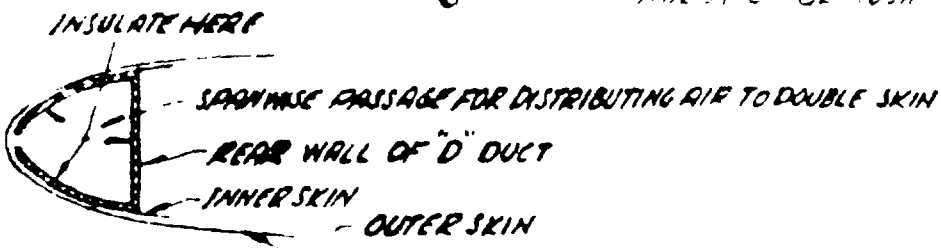
3.6.5 Any insulation which is not naturally impervious to combustible fluids and is used in areas where combustible fluids are present shall be covered and sealed with abrasion- and fluid-resistant covering, preferably stainless steel or aluminum foil.

3.6.6 All ducts and connections used shall be structurally adequate to resist without rupture internal pressures, positive and negative, up to two and one-half times the maximum working pressure at the maximum working temperature. They shall not be subject to permanent deformation when exposed to the normal working pressure and temperature.

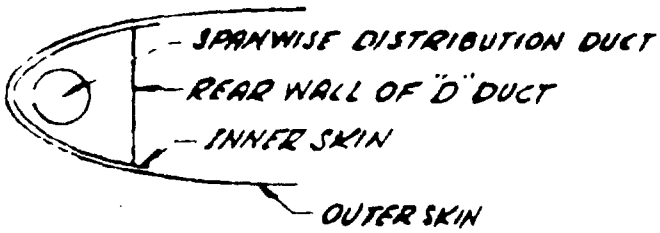
3.6.7 Unless otherwise specified by a component specification, all components of the anti-icing system shall have a minimum service life of 500 hours of system operation.

3.6.8 All parts which require periodic inspection or replacement shall be accessible for inspection and readily removable to facilitate replacement. Quick attachment fittings shall be used for both duct connections and equipment attachments whenever feasible.

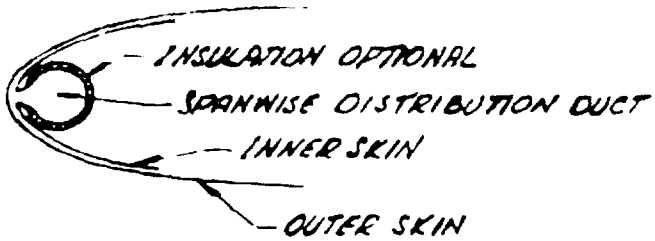
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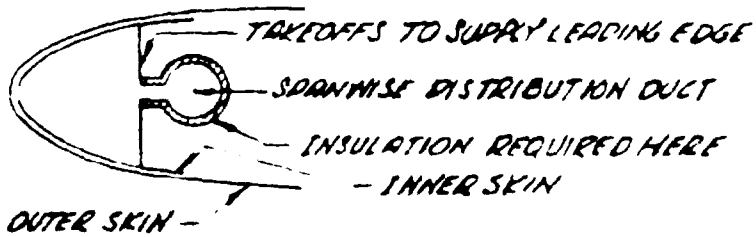
(a) IF NORMAL "D" DUCT CONFIGURATION SHOWN ABOVE IS USED, INSULATION IS REQUIRED AS SHOWN.



(b) INSULATION NOT NORMALLY REQUIRED FOR THIS CONFIGURATION.



(c) INSULATION NOT NORMALLY REQUIRED FOR THIS CONFIGURATION.



(d) WHERE SPANWISE DISTRIBUTION DUCTS ARE INSTALLED AFT OF "D" DUCT AS SHOWN ABOVE, INSULATION IS REQUIRED AS INDICATED.

INSULATION REQUIREMENTS FOR SPANWISE FLOW PASSAGES

FIGURE 2

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3.6.9 If combustion heaters are used as a heat source, they shall be provided with adequate drains which extend beyond the skin line of the aircraft to make certain that all drainage clears the aircraft. Drains which may carry exhaust gases shall not be connected with other drain lines.

3.6.10 Fuel lines or heater accessories containing combustible fluids shall be separated as far as possible from electrical wiring or hot exhaust ducts and in no case may they be installed above heater exhausts or electric wiring. Combustion heaters shall normally obtain their fuel from the main engine fuel-supply system. The use of separate heater fuel pumps shall be subject to the approval of the procuring activity. The heater fuel system shall be provided with a normally closed, electrically operated solenoid shutoff-valve located at the point of takeoff from the main engine fuel system. A normally closed, electrically operated solenoid shutoff-valve shall also be provided at each heater. The heater fuel system shall have a strainer capable of removing all particles of foreign matter from the fuel which are large enough to clog heater fuel passages. Excessively fine filtration is not desired as this will increase filter maintenance and malfunctioning.

3.6.11 Heater exhausts shall terminate in rearward-facing exits displaced sufficiently distant from the skin to prevent exhaust products from overheating or corroding the skin.

3.6.12 If heater exhausts are installed in any compartment which also contains combustible fluids under pressure, the exhaust shall be covered with a metallic shroud and sufficient air shall be passed between the exhaust and the shroud to keep the exposed surfaces at least 100°F below the autogenous kingling temperature of the combustible fluid.

3.6.13 Combustion heaters shall be in accordance with Specification MIL-H-5484.

3.6.14 Each section of rigid ducting shall be sustained by at least two insulated supports spaced sufficiently far apart to provide proper support and alignment of the duct.

3.6.15 A temperature indicator shall indicate the temperature of a typical point on each major anti-iced section that has an individual heat supply or control.

3.6.16 An automatic temperature control shall cycle or modulate the system heat supply in response to wing surface temperature if system operation at ambient temperatures above the design ambient-temperatures would otherwise cause structural overheating.

3.6.17 All electrical equipment shall be designed in accordance with Specification MIL-D-9402 and installed in accordance with Specification MIL-E-7563.

3.6.18 All valves, actuators, controls, and other equipment used in the anti-icing system shall be capable of operating over the extremes of temperature and altitude for which the aircraft is designed and shall not be deleteriously affected by exposure to salt-fog, humidity, vibration, fungus, and sand and dust. Any equipment installed in locations where explosive vapors may exist shall be safe for operation in this location.

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3.6.19 Pipe threads shall not be used for any connection that must be disconnected to permit removal or maintenance of any component of the system, except that they may be used to connect drain lines to combustion heaters.

3.7 Reports:

3.7.1 Anti-icing system data.- Prior to the installation of an anti-icing system in an airplane, the airplane manufacturer shall submit to the procuring activity for approval a report on the design of the system showing a complete schematic drawing of the proposed system and complete data on heat requirements and heat available. These data shall be sufficiently detailed to permit evaluation of the accuracy of the methods used and shall contain complete sample calculations for at least one condition. The report shall also contain a word description of system operation. Submission and approval of this report is a prerequisite for any other approvals on the system or equipment. The report may be submitted in sections or chapters if desired.

3.7.2 Equipment specifications.- The manufacturer shall prepare equipment specifications for all items of equipment used in the system which are not covered by detailed specifications of the procuring activity, which are either designed and manufactured specifically for use in this system or which require additional testing to prove their adequacy for use in the system. These specifications shall be submitted for approval not later than the time at which they are submitted to vendors for bids or proposals.

3.7.3 Equipment test report.- Test reports shall be submitted on all items of equipment used in the system to demonstrate compliance with the test requirements of this specification.

3.7.4 Instrumentation report.- Prior to instrumentation of the airplane designated for flight test of the system, the contractor shall submit a specification or report outlining in detail the instrumentation which is proposed for installation, the methods of calibration which will be used, and sufficient description to indicate the location and significance of each item.

3.7.5 System test report.- Prior to conducting ground or flight tests of the system, the contractor shall submit a test program for approval by the procuring activity. This report, which may be combined with the instrumentation report, shall outline the tests to be conducted, data to be obtained, and purpose of the tests.

3.7.6 Evaluation and extrapolation.- At the completion of the flight testing of the system, a complete report on the testing shall be submitted. This report shall include an evaluation and extrapolation of the data to design icing conditions. Final approval of the anti-icing system is contingent on satisfactory submission on this data.

4. QUALITY ASSURANCE PROVISIONS

4.1 Laboratory tests:

4.1.1 A representative item of each item of equipment used in the system shall be tested to indicate suitability of the design for the use intended. These tests shall be in accordance with the manufacturer-approved equipment specifications and shall include life and performance tests, plus the applicable environmental tests in accordance with Specification MIL-E-5272.

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4.2 Prototype tests:

4.2.1 The ground and flight tests specified below shall be conducted on an airplane designated by the procuring activity to demonstrate compliance of the anti-icing equipment with the requirements of this specification. Wherever possible, the contractor is encouraged to conduct these tests on ground mock-ups, if his facilities permit, to preclude damage to the airplane in case of system malfunction and to ensure early detection of deficiencies. Details of testing shall be in accordance with the manufacturer-approved testing specification and shall include the tests listed in 4.2.2 and 4.2.3.

4.2.2 Ground tests.- Ground tests shall be conducted to demonstrate control system operation, temperature indication system operation, operation of overheat warning or control, general security and safety of the system for flight testing, distribution of available airflow, and freedom from overheating and detrimental effects of differential expansion.

4.2.3 Flight tests.- Flight tests shall be conducted to demonstrate control system operation, temperature indication operation, operation of overheat warning or control, freedom from overheating and detrimental effects of differential expansion. In addition, a complete thermal survey shall be conducted to determine the heat available. These tests shall be conducted in dry air unless otherwise specified by the procuring activity, but should be conducted with sufficient detail and accuracy to permit extrapolation of the results to actual icing conditions. Flight conditions shall include at least the following:

(1) Normal takeoff and climb to operating altitude
(2) Normal descent and landing
(3) Level flight at speed for maximum endurance, maximum speed, and an intermediate speed approximately halfway between these speeds at 5,000, 12,000, and 20,000 feet altitudes

4.2.3.1 Instrumentation shall be installed to determine the quantity and temperature of air from each heat source and the temperature and quantity of airflow in all main distribution ducts. At least 3 wing stations on one side of the airplane and 2 stations on one horizontal and one vertical tail surface shall be instrumented to provide a chordwise profile of exterior and inner skin temperatures as well as temperature drop and airflow through the double skin passages. The remaining wing and tail surfaces shall have sufficient instrumentation to ensure that they are receiving the same heat supply as the completely instrumented section. Sufficient structural temperatures shall be measured to ensure that structural overheating does not occur. Shielded thermocouples shall be used for measuring air temperatures in locations where there is a substantial difference between air temperature and the surrounding metal. If there are discontinuities in the heated areas, sufficient temperature measurements shall be made to determine the effect of the heat flow from the heated to the unheated area. Thermocouples used on aircraft to be tested by WADC shall be copper constantan to reduce the affects of corrosion on instrumentation reliability. Surface temperatures and heat supply shall be measured on scoops or other miscellaneous items provided with anti-icing protection.

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4.3 Individual tests

4.3.1 Each anti-icing system installation shall be inspected and tested to determine that the air distribution and system leakage is within allowable limits as specified in the design analysis. In addition, each installation shall be visually inspected to ensure fabrication and assembly in accordance with approved engineering design for the system and the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Intended use.- The anti-icing equipment covered by this specification is intended for use in aircraft to maintain the airframe free of ice, with the exception of the engine-air-induction system and radome.

6.2 Ordering data.- Invitations for bids and the contracts and purchase orders shall include the conditions and requirements for the following:

6.2.1 Ice protection data.- The ice protection data as specified in 3.2.2 shall be the responsibility of the contractor.

6.2.2 Reports.- The data specified in 3.7.1 through 3.7.6 shall be submitted by the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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